At fixed temperature and pressure, enthalpy and entropy compete to determine the thermodynamic direction of change. Temperature sets the weight of the entropy contribution.

1 A worked competition

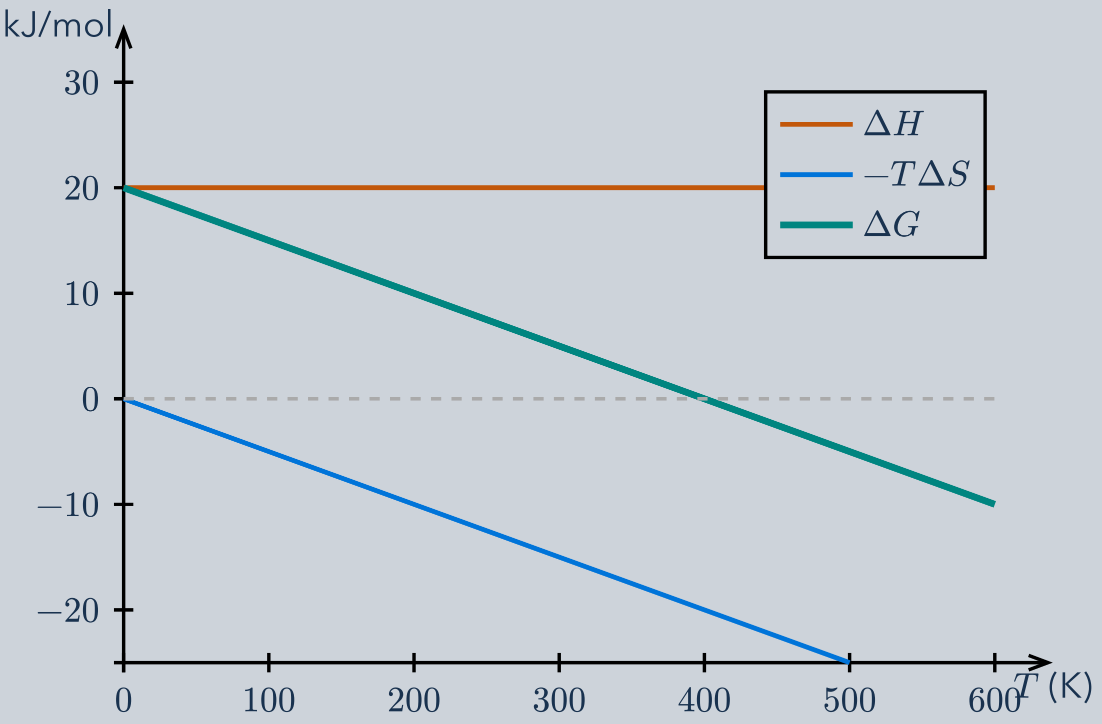
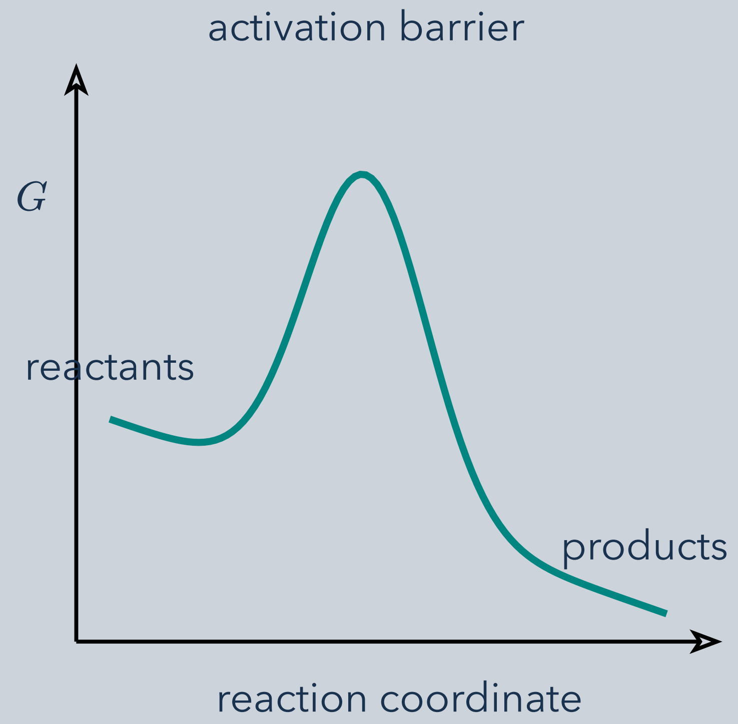


Illustration: $\Delta H = +20$ kJ/mol and $\Delta S = +50$ J/(mol K), treated as constant. Their contributions balance at $T = 400$ K; above it, $\Delta G < 0$.

2 Direction is not speed



A negative free-energy change favors the forward direction at the current composition. A large activation barrier can still make that process very slow.

$\Delta G = \Delta H - T\Delta S$. **Use consistent units.** For a reaction, $\Delta_r G = \Delta_r G^\circ + RT \ln Q$: composition also matters. At equilibrium $\Delta_r G = 0$; a catalyst changes the rate, not this equilibrium condition.